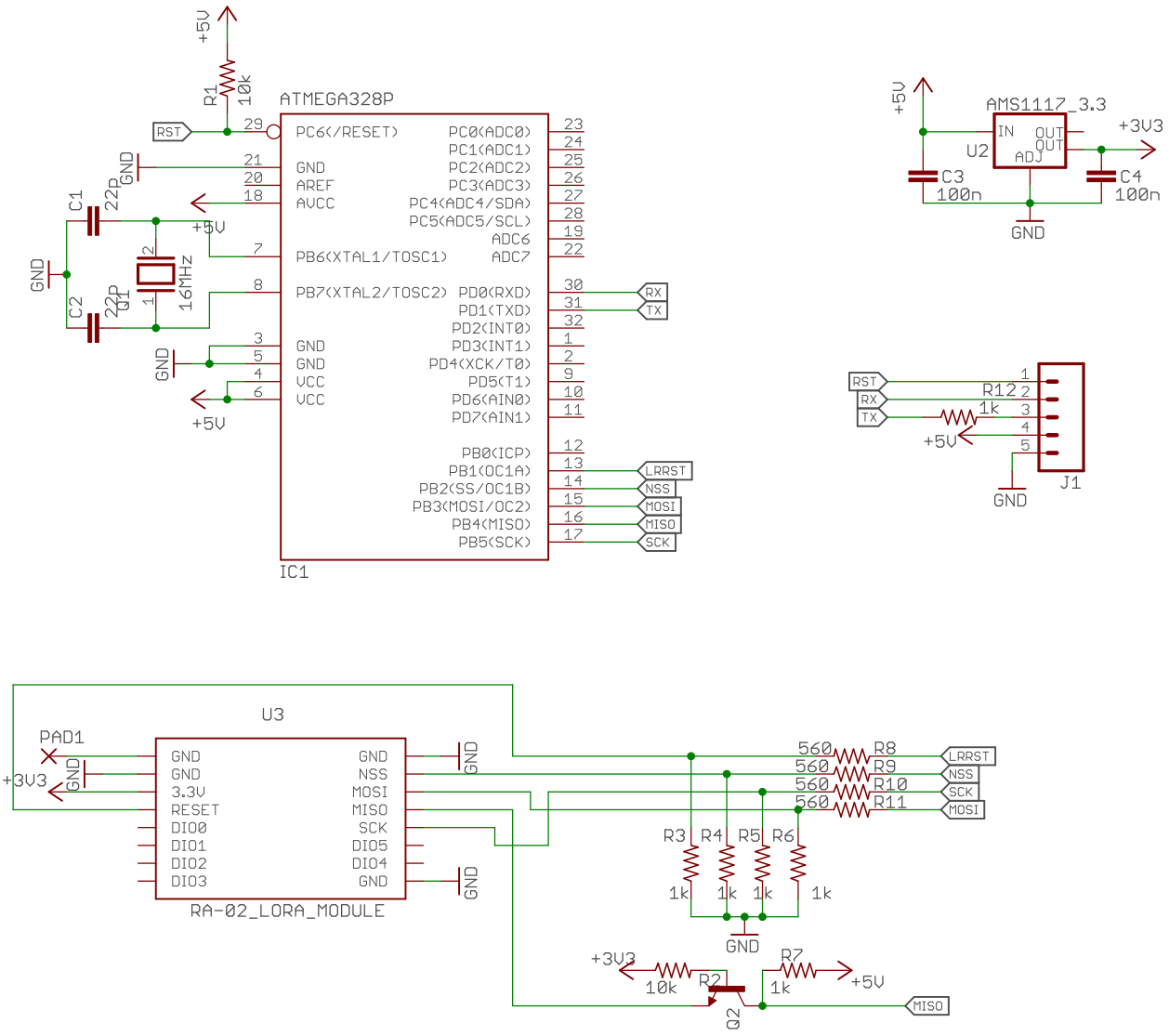
+5V
R1
10k
ATMEGA328P
RST
29
PC6(/RESET)
PC0(ADC0)
23
PC1(ADC1)
24
GND
21
GND
PC2(ADC2)
25
AREF
20
PC3(ADC3)
26
AVCC
18
PC4(ADC4/SDA)
27
PC5(ADC5/SCL)
28
ADC6
19
PB6(XTAL1/TOSC1)
ADC7
22
C1
22P
GND
+5V
Q1
16MHz
12
7
C2
22P
GND
PB7(XTAL2/TOSC2)
PD0(RXD)
30
RX
8
1
PD1(TXD)
31
TX
PD2(INT0)
32
3
GND
PD3(INT1)
1
GND
5
PD4(XCK/T0)
2
4
VCC
PD5(T1)
9
VCC
6
PD6(AIN0)
10
+5V
PD7(AIN1)
11
PB0(ICP)
12
PB1(OC1A)
13
LRRST
PB2(SS/OC1B)
14
NSS
PB3(MOSI/OC2)
15
MOSI
PB4(MISO)
16
MISO
PB5(SCK)
17
SCK
IC1
+5V
AMS1117_3.3
IN
OUT
OUT
ADJ
U2
+3V3
C3
100n
C4
100n
GND
RST
RX
TX
R12
1k
+5V
1
2
3
4
5
J1
GND
U3
PAD1
+3V3
GND
GND
GND
3.3V
RESET
DIO0
DIO1
DIO2
DIO3
GND
NSS
MOSI
MISO
SCK
DIO5
DIO4
GND
GND
RA-02_LORA_MODULE
560
R8
560
R9
560
R10
560
R11
LRRST
NSS
SCK
MOSI
R3
R4
R5
R6
1k
1k
1k
1k
GND
+3V3
10k
R2
Q2
R7
1k
+5V
MISO